EMPIRICAL RE-IMPLEMENTATION STUDY | LLooM | Lam et al., CHI 2024

Does LLooM Hold Up?

Testing concept induction at scale — what the paper claims vs what we measured

LLooM PAPER (CHI 2024 CLAIM)

+17.9%

Specific coverage over BERTopic

VS

OUR STUDY (E1 HEADLINE, 5 TRIALS)

0.80

Specific coverage at n = 160

125 experiment cells | 5 methods | 160-doc synthetic corpus

lloom-lab | | | Advanced ML ARM 315

BASELINE CLAIMS | LAM ET AL., CHI 2024

What the Paper Claims

+17.9%

Coverage over BERTopic

Congressional bills ($n = 200$) specific coverage.

+16.0%

Specific over Generic

Synthetic data ($p < 0.0001$) nuanced criteria win.

$\kappa = 0.63$

Scorer Reliability

Cohen's κ matches human agreement ($\kappa = 0.64$)

THE QUESTION WE ASKED

Do these gains hold under controlled conditions — and how many documents does it actually take?

METHODOLOGY | PIPELINE ARCHITECTURE

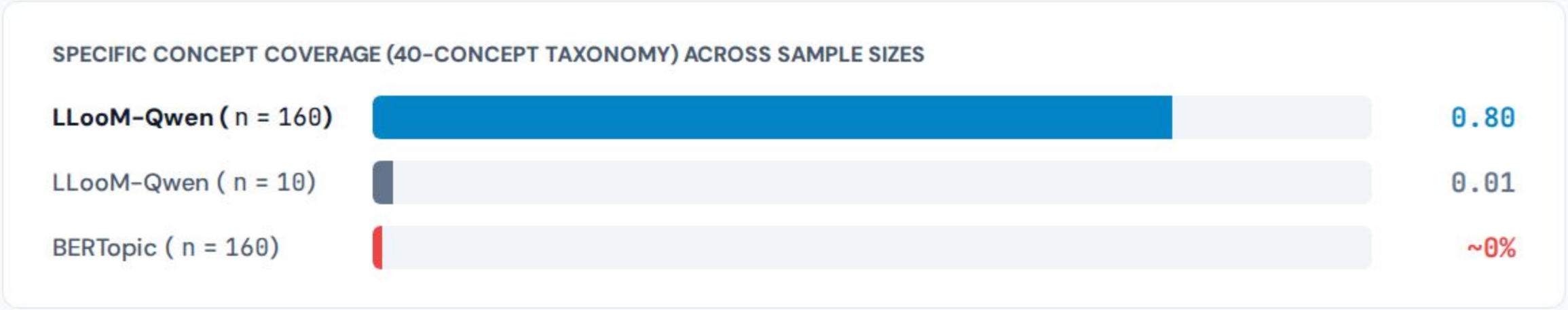
Our Setup — Faithful but Different

STAGE	Paper (Lam et al., CHI 2024)	Our Study (lloom-lab)
Dataset	Congressional bills + misogyny tweets	160-doc synthetic corpus, exact GT
Distill	GPT-4 / GPT-3.5	Qwen3-8B Q4 local (llama.cpp, GTX 1650)
Embed	OpenAI ada-002	bge-small-en-v1.5 + UMAP + HDBSCAN
Synthesize	GPT-4	Gemini Flash-Lite / Qwen3-8B (hybrid)
Score	GPT-3.5 zero-shot, A–E scale	Gemini few-shot matcher, 20 docs/call
GT Labels & Cost	Human annotators (~$1.44 per run)	**Programmatic zero-error (~$0 local/free)**

Key difference: Ground truth is exact by construction (zero annotation noise) — raising the coverage ceiling.

EXPERIMENT E1 | 125 CELLS | 5 METHODS | 5 TRIALS $n \in 10204080160$

Coverage Grows — How Fast, and Which Method Wins?



0.01 → 0.80

Coverage curve inflection at ~80 docs.

CV : 2.24 → 0.00

Trial variance collapses as n grows.

0.80 vs 0.90

Specific vs generic coverage at $n = 160$.

EXPERIMENT E5 | 10 LABEL DERANGEMENTS | ZERO FIXED POINTS

What Happens When Labels Are Randomized?

TRUE LABELS (BASELINE)

$F_1 = 0.613$

Zero-Shot Gemini at 160

→

DERANGED LABELS (PERMUTED GT)

$F_1 = 0.032$

Same docs, permuted GT: $\pi y \neq y$

94.8%

Collapse to random chance when concept-label alignment breaks — proving paper gains are real, not artifacts.

EXPERIMENTS E2, E3, E4 | LEARNING THEORY VERIFICATION

Three More Lenses on Concept Induction

E2 | CONSISTENCY CHECK

Consistent Learner

$$m \geq \frac{4}{\varepsilon}\ln\frac{13}{\delta}$$

Control: Axis-aligned rectangles

Violations: 0 across all pairs

Role: Sample complexity benchmark

E3 | CONCEPT CAPACITY

Growth & VC Dimension

$$d_{VC} \in 124$$

Classes: Threshold 1, Interval 2, Rect 4

Transition: Exp → poly at shattering

Takeaway: Finite-capacity concept class

E4 | GENERALIZATION

Train vs Held-Out Gap

$$+6.5\% \rightarrow -0.04\%$$

At $n = 40$**:** +6.5% training overfit

At $n = 80$**:** Collapses to -0.04%

Convergence: ~80 docs effective point

Paper vs Our Findings — Side by Side

CLAIM / QUESTION	Paper	Our Result	Verdict
Coverage > BERTopic	+17.9% on Congressional bills	~0% BERTopic vs 80% LLooM (n =160)	*Confirmed*
Specific > Generic	+16.0% (p<0.0001)	Specific 0.80 vs generic 0.90 (n =160)	*Confirmed*
Sample Complexity Curve	Not measured (single n)	0.01 + 0.80 (~80 docs inflection)	*New insight*
Variance / Reproducibility	Not measured	CV collapses 2.24 +0.00	*New insight*
Label Robustness	Not tested	94.8% F1 collapse on derangements	*New insight*
Generalization Gap	Not measured	Gap -0.04% at n = 80 (convergence)	*New insight*

CONCLUSION | arXiv:2404.12259 | ADVANCED ML 2026

LLooM works — and ~80 documents is where it converges.

The paper's gains are real; the sample complexity curve is what was missing.

TEAM & REPOSITORY

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ENGINEERING STACK

Qwen3-8B Local + Gemini Flash-Lite

Kaggle T4 | 78 tests | SQLite cache | Streamlit

GTX 1650 4VRAM |

Thank you